

Compact Representations of Human Interception Movements

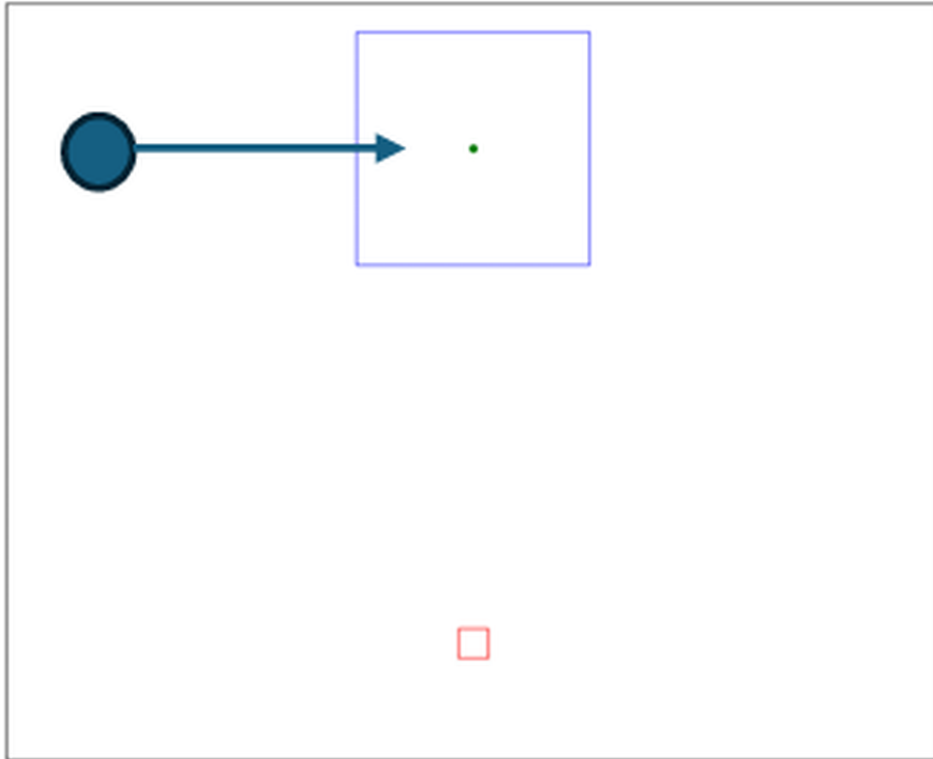
Low-dimensional models for representing and generating human interception movement

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10-minute presentation and live demo



The experimental task

- **Participant's task:** move the finger from the lower start zone into the upper interception square exactly as the ball passes through it.
- **Conditions:** the free eye-movement condition is analysed. Six target start locations (3 right, 3 left) over a continuous speed range.
- **Sample:** 4,732 valid trials from 28 participants, measured in 3D at 240 Hz with a Polhemus system.

A strict split that prevents information leakage

Four fixed folds: 17 training participants, 4 validation and 7 test. Every participant appears in a test fold exactly once.

What we set out to answer

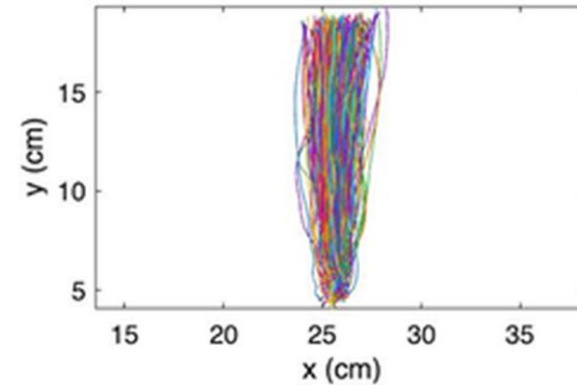
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The core question

Human movement is stochastic: perception and motor control vary from trial to trial. One participant, one target - and every trial takes a different path.

Can a compact fingerprint capture that spread?

Compress 200 numbers - the 100×2 path - into a 3-to-8 dimensional fingerprint of an individual's movement distribution.



100×2 coordinates $\longrightarrow$ $n = 3$ or 8 latent numbers (durations predicted)

What would count as success, evaluated on participants the model never trained on

SUCCESS TEST 1

Reconstruct

Rebuild specific recorded trajectories from the compressed code.

SUCCESS TEST 2

Generate

Sample new trajectories matching that person's stochastic distribution.

SUCCESS TEST 3

Stay personal

The fit must come from that person's own fingerprint, not a generic one.

Trajectory processing and the compression architectures

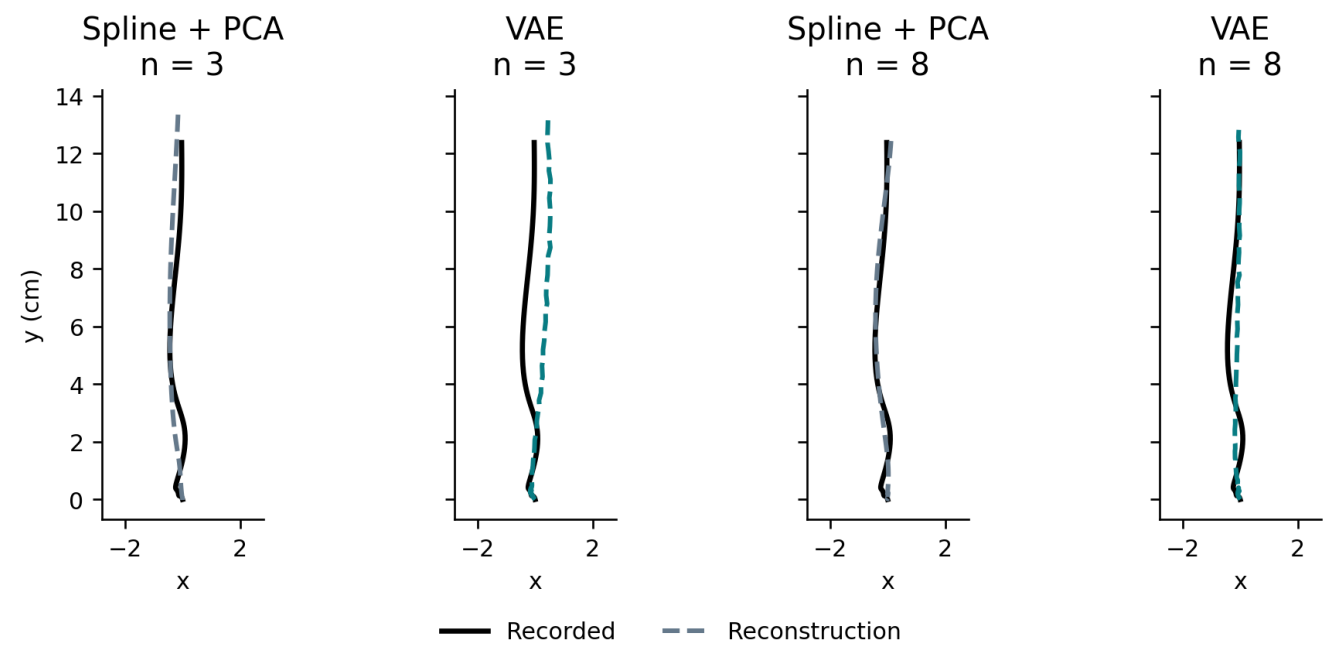
Stage in the model	Spline + PCA (linear baseline)	Neural models (VAE / CVAE / CAE)
Primary input	The same 100×2 normalised trajectory	The same 100×2 trajectory; CVAE and CAE add 5 task conditions
Where standardisation happens	On the 18 spline coefficients, after fitting	On the 200 coordinates, with train-set constants
Encoder / compression	Cubic B-spline fit, 9 coefficients per axis -> PCA to n latents	Two fully connected 256-unit ReLU layers -> latent n
Decoder / reconstruction	Inverse PCA -> multiply by the spline basis for the full path	Two 256-unit ReLU layers -> separate heads for 200 coordinates and 2 durations
Timing prediction	Separate Ridge regression on the PCA scores + conditions	Dedicated head predicting initiation and movement time

Three neural variants, one architecture

VAE - variational autoencoder, trained on the trajectory alone. CVAE - conditional variational autoencoder, which also receives the task conditions. CAE - conditional autoencoder, the same network without the variational latent.

All are compared at identical latent dimensions, $n = 3$ (extreme compression) and $n = 8$ (extended capacity).

Task 1: Trajectory Reconstruction



One median-duration trial. Black: recorded; dashed: reconstructed. Both axes in cm at equal scale.

Trajectory reconstruction error (MSE, cm²)

Model	MSE n=3	MSE n=8
Spline + PCA	0.0972	0.0228
VAE (Unconditional)	0.1483	0.0335
Conditional AE (CAE)	0.1697	0.0260
CVAE	0.1770	0.0325
Condition-only Ridge (Floor)	1.2338	1.2338

Key insight on pointwise reconstruction

PCA wins at reconstructing a single trajectory.

As a linear projection, PCA is analytically optimal for minimising squared error among all linear methods. VAEs trade off pointwise accuracy to obtain a continuous and samplable latent space.

Task 2: generating a personal movement distribution

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How do we generate movements from a fingerprint?

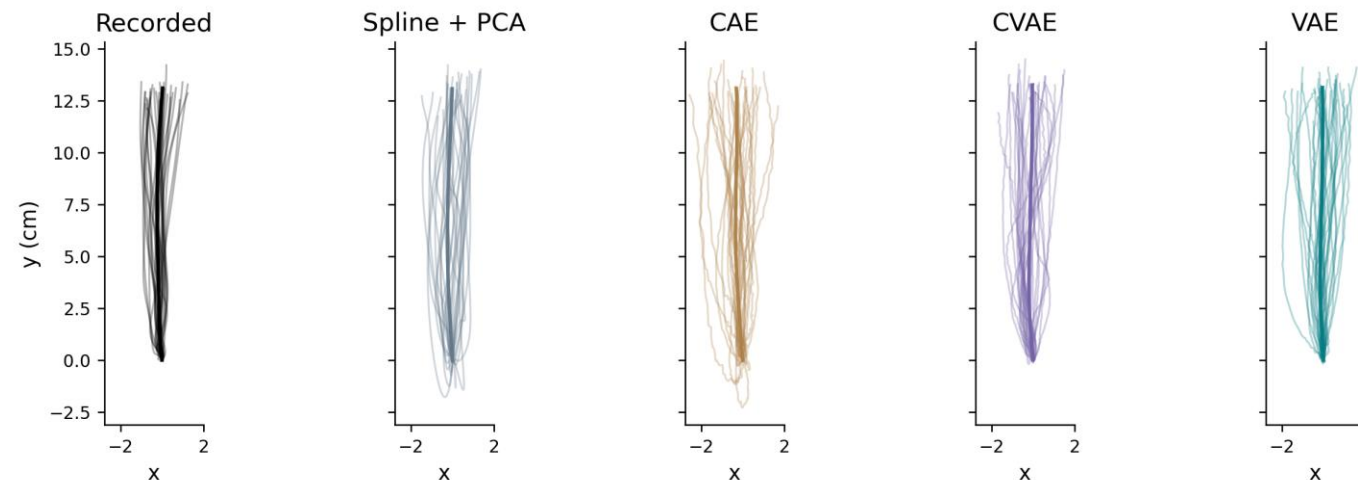
- **Enrollment:** earlier context trials of the participant, whose mean defines the personal fingerprint centre.
- **Stochastic sampling:** draw 120 latent vectors around that centre using a shared covariance learned from training data.
- **Unseen movements:** query trials are held aside to evaluate the distribution and are never exposed to the model.

Below, $n = 8$. Thin: first 30 draws. Thick: their mean. Equal x/y scale, cm.

$$\text{fingerprint centre: } \theta_s = (1 / |C_s|) \sum c_i$$

$$\text{one generated movement: } z \sim N(\theta_s, \Sigma_{\text{train}})$$

c_i is the trial's code: the encoder mean μ_i for the neural families, the PCA scores for spline + PCA.



WHAT WE MEASURE

11 kinematic features

Initiation and movement time, peak speed, time to peak, path length, straight-line distance, curvature, maximum lateral deviation and the three endpoint coordinates.

HOW WE COMPARE

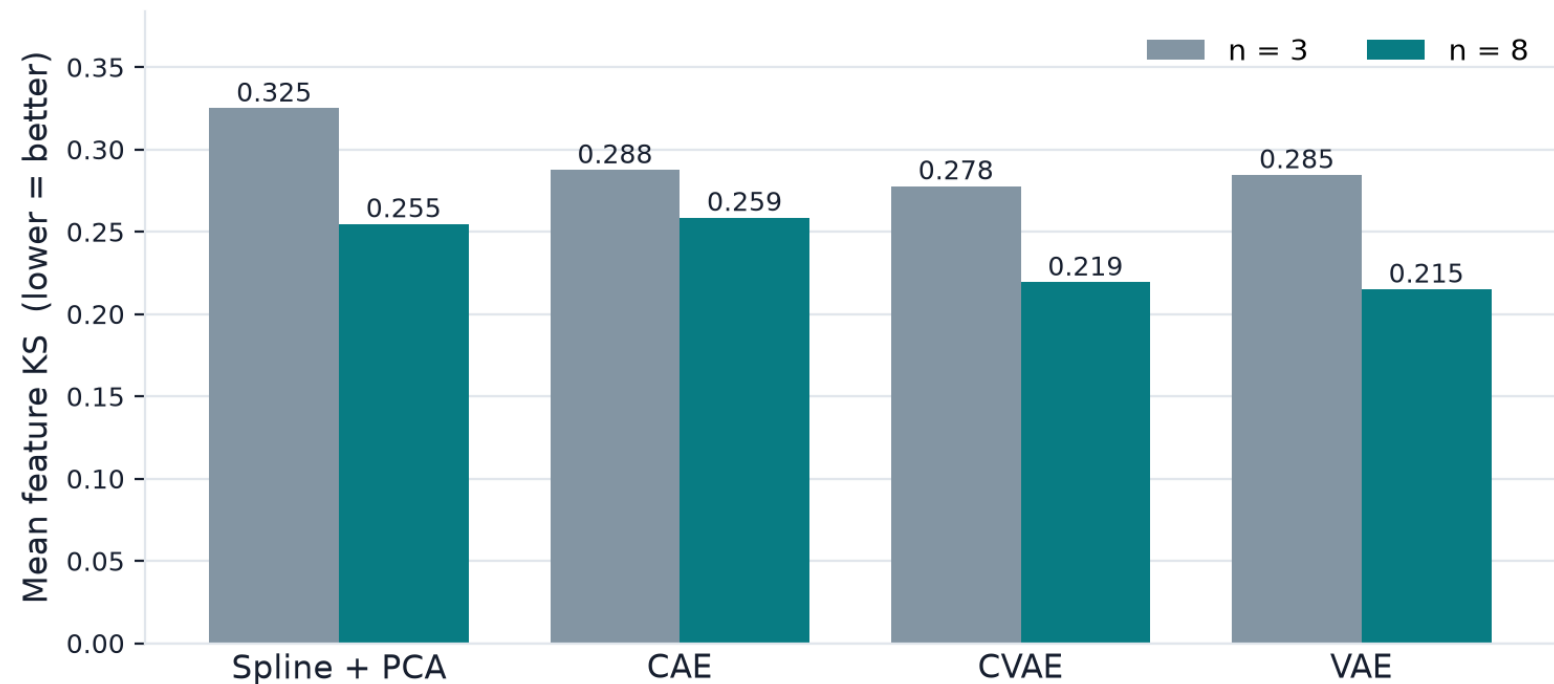
Two views of the distribution

One KS test per feature, averaged over the 11 into Mean KS. Energy discrepancy and MMD squared judge the joint structure of all 11 at once. Lower is better throughout.

One fixed yardstick for every model

Feature scales and the RBF bandwidth are fitted on training participants only and then frozen, so every model is measured against the same geometry rather than one tuned to its own output.

Generation results: VAE leads on movement distributions



A reversal in the ranking: VAE wins at generation

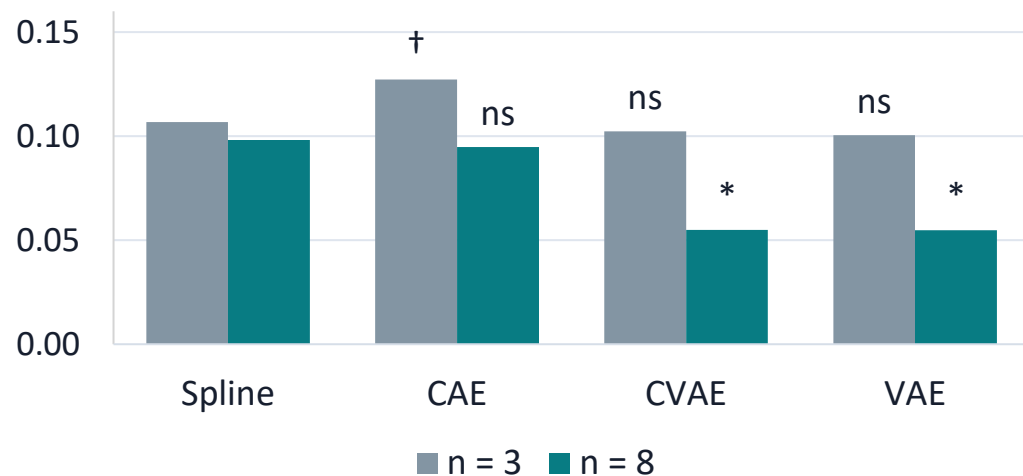
Although Spline + PCA won pointwise reconstruction, VAE and CVAE achieve much better distribution matching (corrected $p < 0.001$ under BH and Holm). VAE at $n = 8$ was set as the dashboard default.

The two joint measures agree: at $n = 8$, VAE reaches energy 0.4110 and MMD squared 0.0639, against 0.7347 and 0.1153 for Spline + PCA.

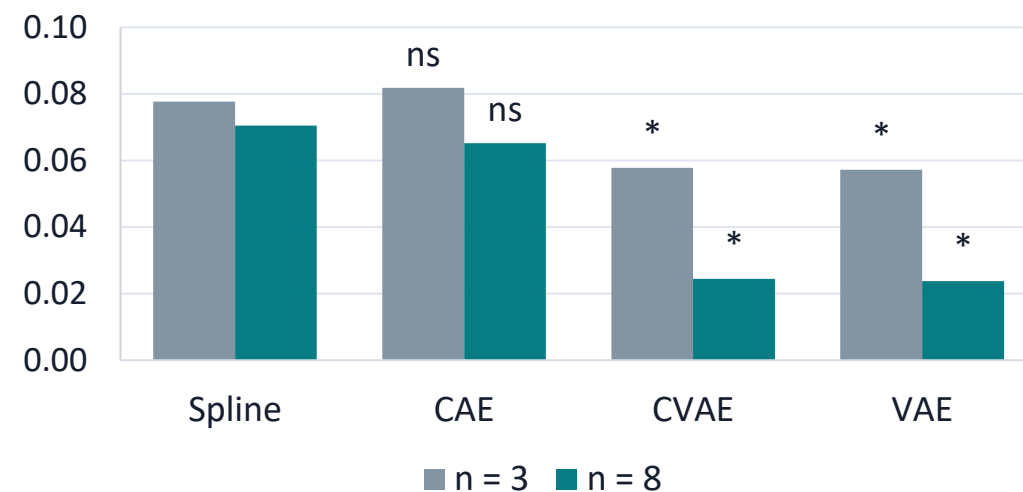
Do generated paths match the held-out set of complete trajectories?

Energy and MMD² compare between-set differences with within-set variability. Lower is better.

Full-path energy (cm)



Full-path MMD squared



Vs spline, same n: * BH + Holm < .05 † BH only ns neither | 28 participants; paired tests, 80 comparisons

At n=8, VAE and CVAE outperform spline + PCA on both distances

The finding also survives balancing the x/y axes. VAE and CVAE are not significantly different; n=3 evidence is mixed.

Personal context improves generation

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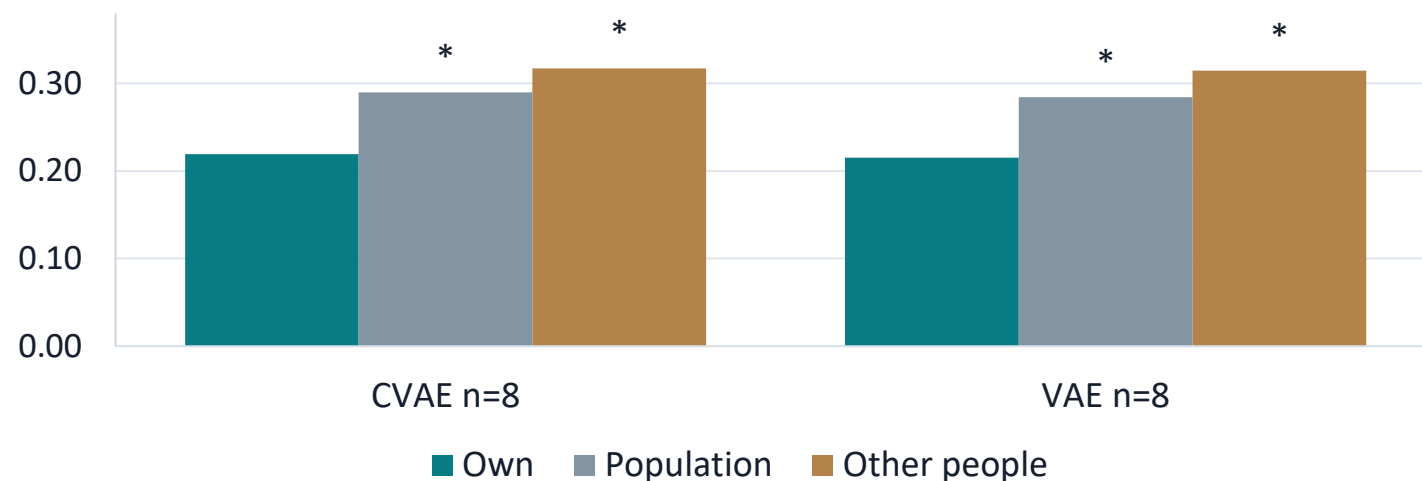
Replace only the fingerprint center; keep the decoder, noise and conditions fixed.

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participants had lower mean KS

Own center vs population center
VAE, n=8

Mean feature KS (lower is better)



* Different from own center: BH-adjusted $p < .05$

The correct personal center improves the generated feature distributions

Both n=8 models improve on population and other-person centers. This supports useful personal information within the session.

An interactive dashboard

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Explore a latent representation and see its generated trajectory and predicted timing.

